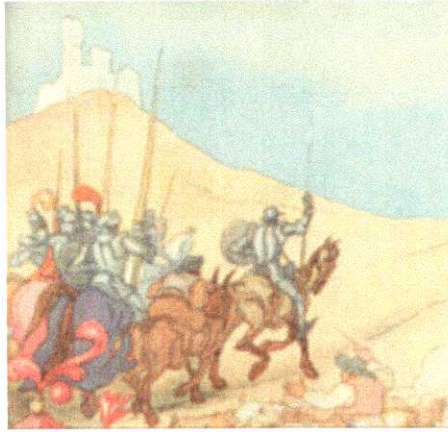




Symposium on

**The Potential for Interstellar Travel and
Unidentified Aerial Phenomena**

2002 November 8

Human Beings in the Galaxy:
.... or, Babes in the Woods

Professor Richard Conn Henry
The Johns Hopkins University

We human beings, probably have been in existence as a “speaking species” for only perhaps the last 100,000 years, out of Earth’s five billion year history.



And, our species has emerged to *dominate this planet*, only over the last 10,000 years.



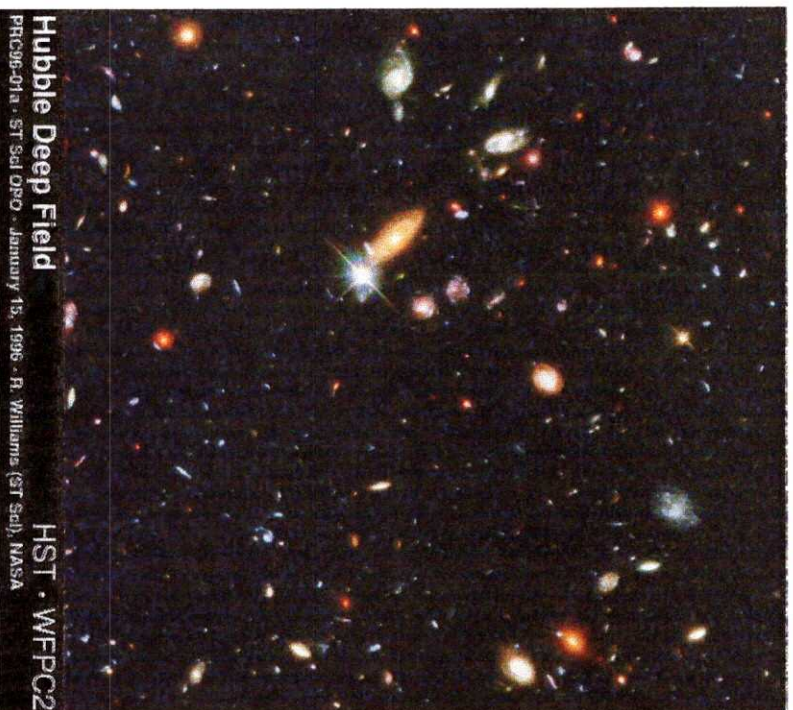


We humans have had access to *serious* physics (**Principia**) for just 315 years,

$$H|\psi\rangle = i\hbar|\dot{\psi}\rangle$$

and we have had *correct* physics (**quantum mechanics**) for less than 100 years.

The **Copernican Revolution** showed us that *our Earth* is merely a *minute portion* of an **inconceivably vast universe**; and *seemed*, at least, to reveal to us that we are only an *utterly unimportant* element in that truly humungous universe.

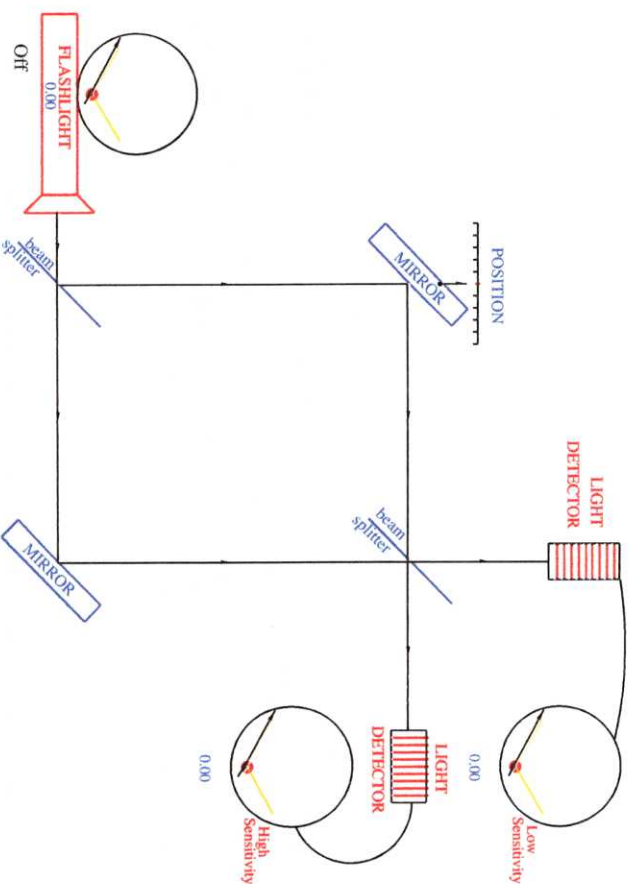


So! We have discovered that **we humans are “newborn babes”** and that we are *on our own* in an utterly vast wilderness.

Should we not be both shocked and terrified?



Copernican Revolution



Quantum Mechanics, in the *orthodox* Copenhagen Interpretation, has now shown us that (in the words of Eddington), “the stuff of the world is mind stuff.”

Entirely unexpected!

“Measurements Are the Only Reality, Say Quantum Tests”
(Science Magazine).

Our attention therefore goes to the **Big Question**: *How many minds are there* in this inconceivably vast universe?

Well, I *know* there is *my* mind. I concede *your* mind! And, *other*

people; and, dogs and cats ... and oysters?

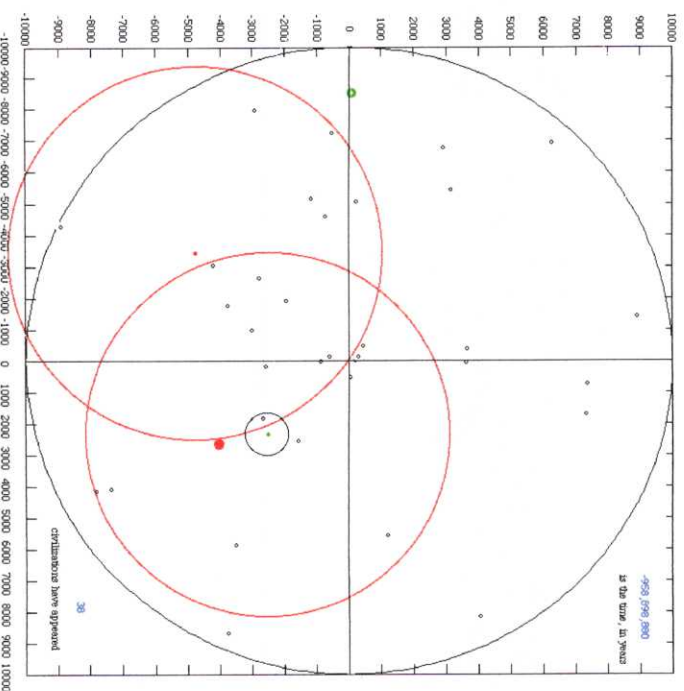
I have heard Native Americans say “we Native Americans believe that a *stone* is *alive*,” and—quantum mechanics has forced me to listen to them *with interest*.



Are there *other* minds that are “out there,” in our own galaxy of 200,000,000,000 stars? and/or, in *any* of the *billions* of such galaxies we have discovered?

Enrico Fermi famously asked, “**Where Are They?**” in reference to such other civilizations.

Fermi had recognized that even with conventional, known physics, a species such as ours should be able to colonize the entire galaxy in a mere 20 to 40 million years, a number that is very *small* compared with the perhaps 12 billion year age of our galaxy (the Big Bang having occurred 14 billion years ago).

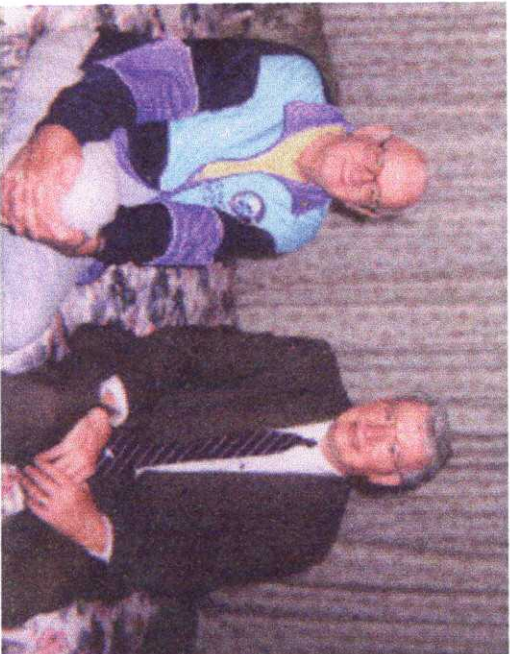


Now, the chance of there being another “newborn” civilization, similar to ours, here in our galaxy, today, is *negligible*. Just do the numbers!



No, if there *are* other civilizations in our galaxy, those civilizations are **tens to hundreds of millions of years ahead of us** (in terms of technology and scientific achievement).

Any such civilization could “run rings around us,” and *no one can know that they are not doing precisely that*. No one.



“Sufficiently advanced technology is indistinguishable from magic.” — **Sir Arthur C. Clarke**

But any such civilizations should surely have *colonized the galaxy*, yet there is *no evidence* that such colonization has occurred.

Perhaps ... we are really ... utterly alone. Perhaps our destiny is to colonize the galaxy ourselves.

Yet, I find it *difficult to believe* that what is happening now, here on Earth, has not happened elsewhere in our galaxy; and many times, and of course ... long ago.

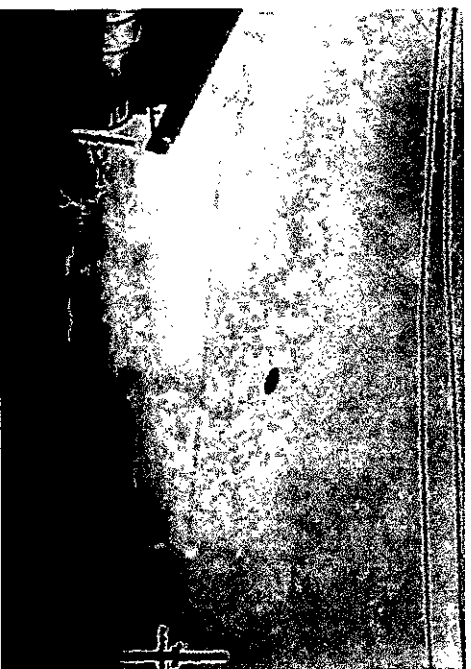


The reported behavior of UFO's is *incompatible* with my anticipation of how such a civilization would behave. *I feel like issuing them with a citation!*



That summarizes my position regarding life in the universe, and the reality of UFO's.

So, now, I come to my *conclusions*. I will, of course, be *conservative*: it would be foolish *not to be* conservative.



First of all, I put on my hat as a *scientist*. Use of Occam's razor allows me quickly to conclude that hoaxes and mental problems are perhaps more likely to account for the bizarre UFO reports than are visits from alien civilizations. So I conclude, conservatively, *as a scientist*, that we are not being visited, and have not been visited, by alien civilizations.

Notice how *easy* it was for me to come to that conclusion, as a scientist! It really does show the *power* of Occam's razor!

Notice, in particular, that **I did not even need to examine any evidence** that may exist.

My training is as an *astrophysicist*. I know *little or nothing* about hoaxes; I know *little or nothing* about mental problems. And, like **every** human being *whatever their education and career choice*, I know **nothing whatsoever** about highly advanced civilizations.

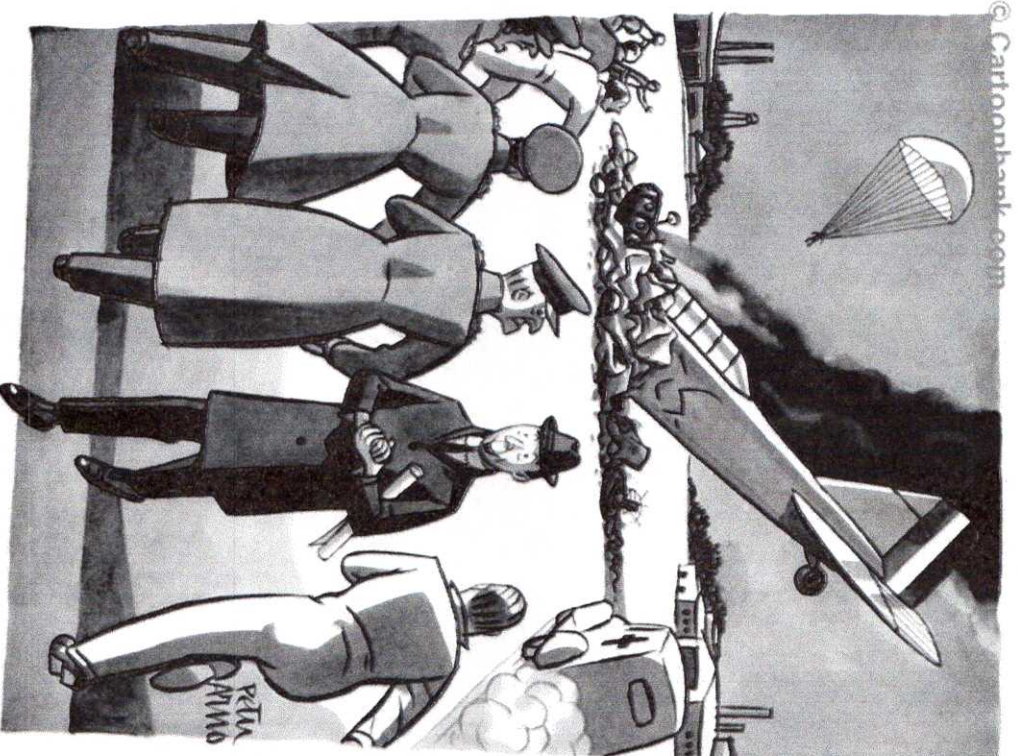
Yet, I can come to a *conservative* conclusion that is in accord with the scientific standard of Occam's razor! Isaac Newton: "**We are to admit no more causes of natural things than such as are both true and sufficient to explain their appearances.**"

Of course, *I could be wrong.*

You see, —
being wrong is a *normal* part
of the scientific process.

*“Well, back to the old
drawing board!”*

My point is, that the *scientific*
standard is **not** the standard
that it is *wise* to apply, when
considering the phenomenon
of UFOs.



“Well, back to the old drawing board.”

For me, the matter must not *end* with Occam's razor, for my hat as a scientist is not the *only* hat that I wear.



Now, instead, let me put on my hat *as a simple person living on this Earth*, and I recognize: that we are **newborns, living in one humungous universe**. A universe where, clearly, there *could well be* alien civilizations, which perfectly well *could be* visiting us, and for purposes that *I could never conceive*. And, I recognize that, since that possibility clearly does exist, then *we should be alive to it! Ignoring it would be the very opposite of being conservative!*

The good news is that *being alive to it* is *not* an expensive proposition. We are already investigating the universe, scientifically, with a wonderful NASA program of space astronomy. **To also systematically and carefully investigate and assess potential evidence of the presence of alien intelligence is a comparatively minor additional expense.** And, would be a wise *insurance policy*.



Collect and collate evidence of anomalous phenomena! What kind of a civilization would not collect and collate evidence of anomalous phenomena? Only a foolish and short-sighted civilization indeed!



justice boy,

short-sighted civilization indeed!

Our English and



Did the Aztecs have a
“**National Aquatics and
Seafaring Administration,**”
that, in addition to such
activities as building boats,
and recording the ocean’s
tides, was wise enough to
*catalog and collate and
analyze sightings of*
“**Unidentified Floating
Objects,**” or UFOs?

Well, the Aztecs *did not*! But: *they should have*! And, they
should *not* have ridiculed such things as, sightings of creatures
with two heads and four legs!

We must *do better* than the Aztecs did. *To fail to vigorously investigate anomalous phenomena, would be short-sighted and foolish.*

Some of the other speakers will draw your attention to anomalous phenomena that I believe deserve thorough investigation, *whatever* their origin.

And, we are foolish indeed, if we do *not* search for extraterrestrial intelligence, here at home; in our galaxy; and in the universe.

I do hope that we will sharply increase our efforts in these regards, over coming years.

Thank you for your attention.